Spring 2025 Homework Modeling Exercises Weeks 3 & 5

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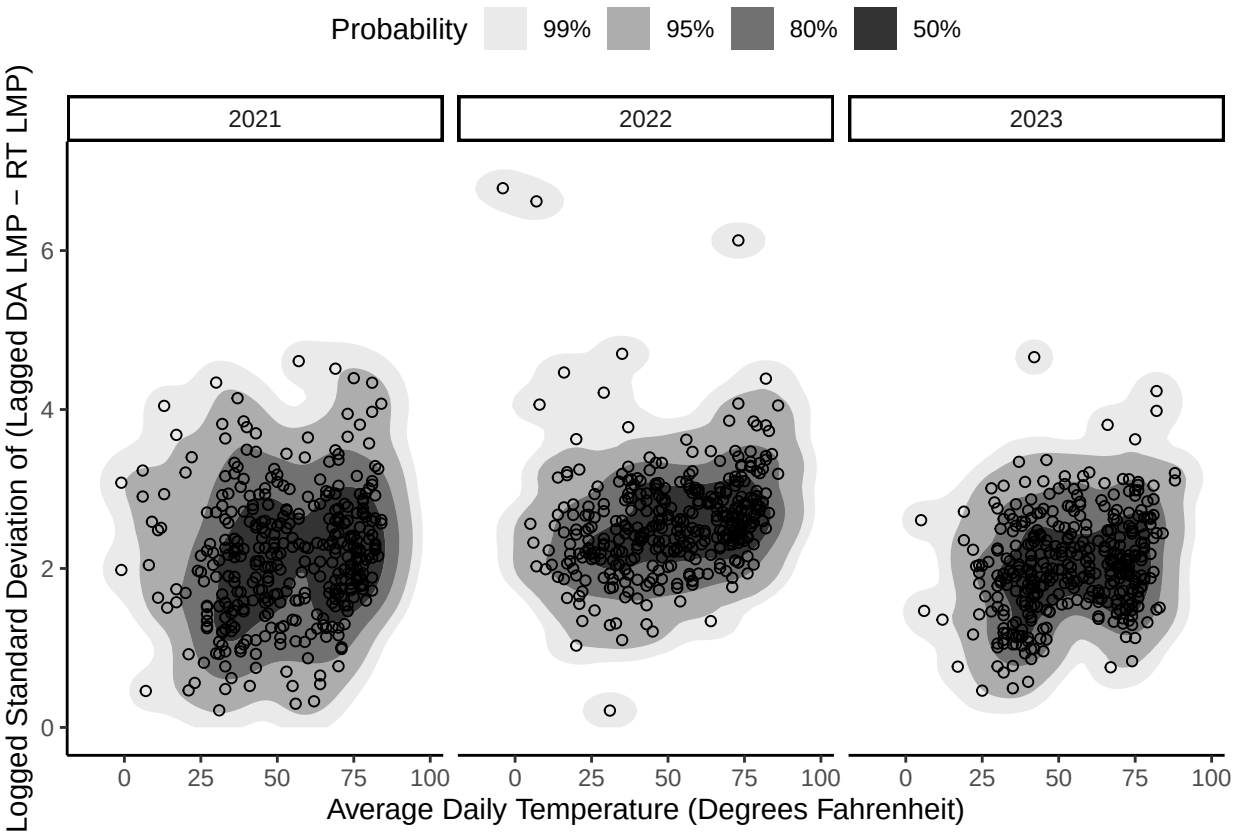
25. The code below plots the joint density of average daily temperature and the logged standard deviation of the daily volatility of the RT-DA LMP spread. Is there an obvious pattern thatemerges? Try identifying notable outliers in the lmp_temp_Chicago dataset. Conjecture as towhy outliers emerged on those days.

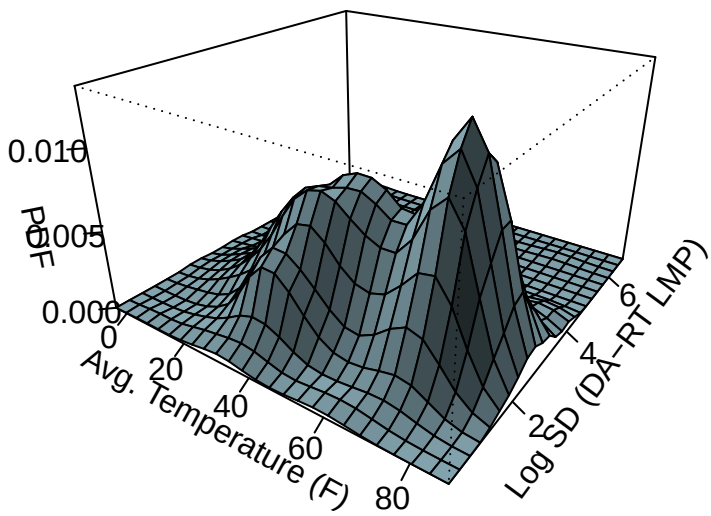
```
ggplot(lmp_temp_Chicago, aes(x=temp, y=log(vol_spread)) ) +geom_hdr(xlim = c(-25, 100), ylim = c(0,10)) +geom_po
```



26. The code below renders the plots above using a 3D density plot. Give it a try! What do thepeaks and valleys represent?

```
dens_Chicago <- kde2d(
  lmp_temp_Chicago[date > '2021-01-01']$temp,
  log(lmp_temp_Chicago[date > '2021-01-01']$vol_spread)
)

persp(dens_Chicago$x, dens_Chicago$y, dens_Chicago$z,
      xlab = "Avg. Temperature (F)",
      ylab = "Log SD (DA-RT LMP)",
      zlab = "PDF",
      theta = 35, phi = 25, expand = 0.6,
      col = "lightblue", shade = 0.4, ticktype = "detailed",
      main = "Joint Density: Temperature vs. Spread Volatility")
```

Joint Density: Temperature vs. Spread Volatility

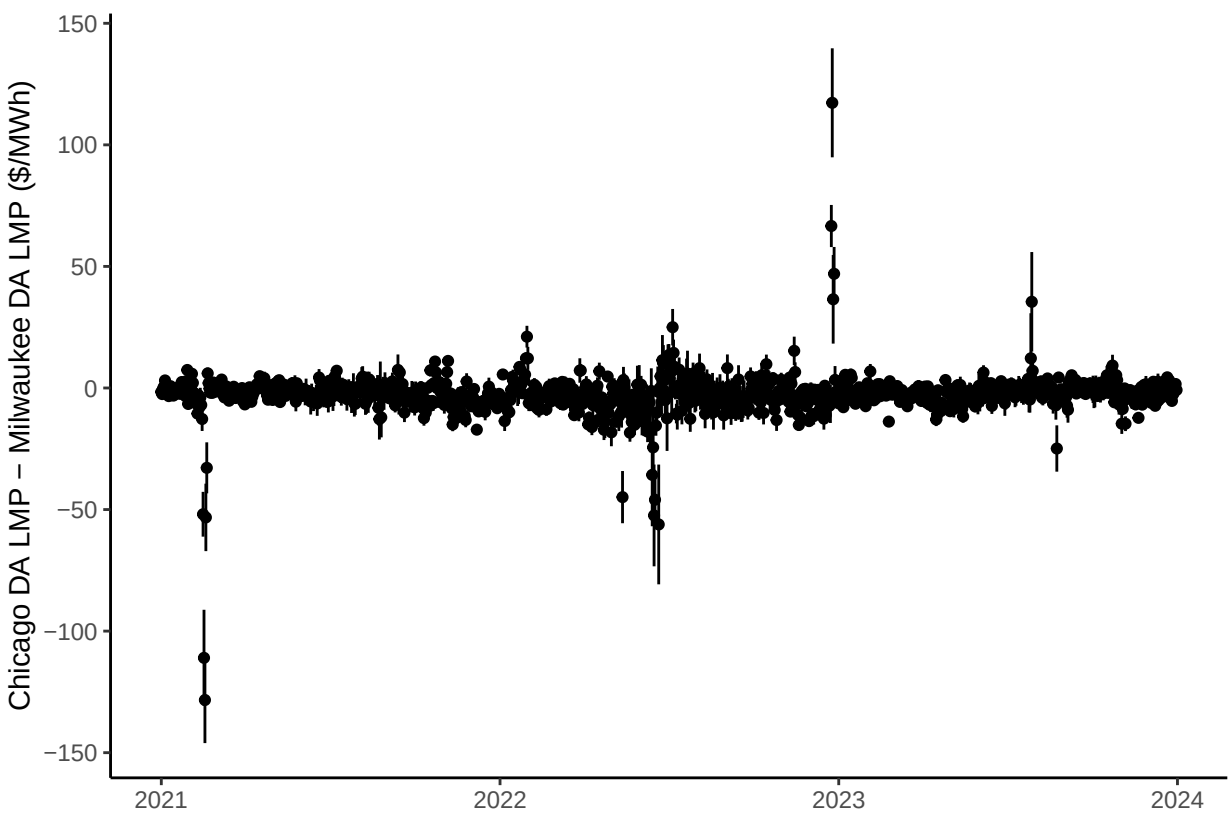


VII. Comparing Interconnected Transmission Regions

The states west of PJM belong to the Midcontinent Independent System Operator (MISO). Milwaukee and Chicago, two cities merely 90 minutes apart by car, belong to two different ISOs(Milwaukee in MISO and Chicago in PJM). In this section, we explore differences in LMPs and their responsiveness to weather patterns, in two separate (yet interconnected) parts of the grid.

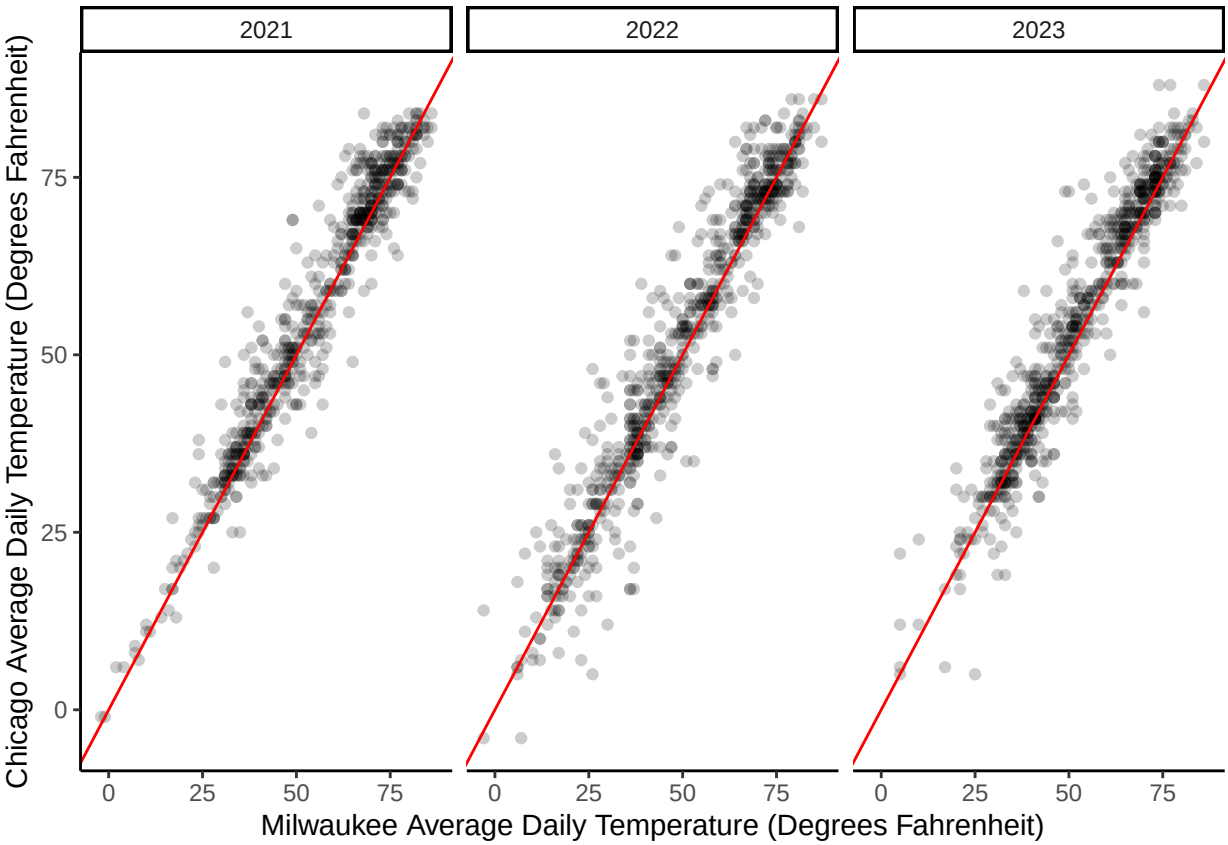
27. The code below plots the spread between average daily DA LMPs in Chicago and Oak Creek Power Plant (near Milwaukee). Consider the outlier days. Could they be explained byvariations in weather? Plot the same figure for the spread in average daily temperatures totest your hypothesis.

```
lmp_cities <- data.table(date = data_Milwaukee$date,Milwaukee = data_Milwaukee$DA_LMP,Chicago = data_Chicago$tot

ggplot(lmp_cities, aes(x=date,y=da_spread)) +stat_summary(size=0.2) +labs(x='',y='Chicago DA LMP - Milwaukee DA
```



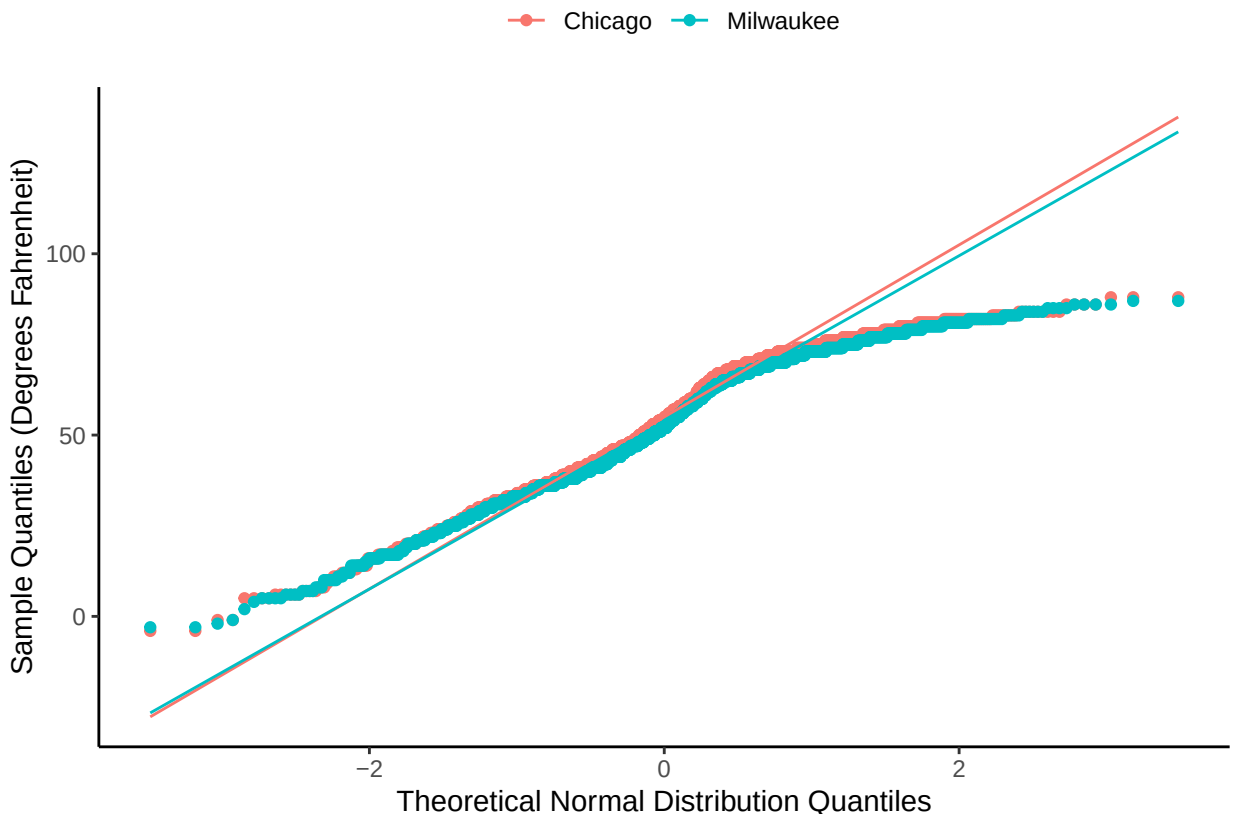
28. One might expect that weather in Milwaukee and Chicago to be similar. The code below generates scatter plots comparing average daily temperatures in Milwaukee and Chicago, showing temperatures in the two cities are closely aligned. Try creating a similar plot foraverage daily DA LMPs in the two cities. How closely aligned are the average daily DALMPs in the two cities? Why might LMPs in the two cities differ more than temperatures

```
temperature_cities <- data.table(
  date      = data_Milwaukee$date,
  Milwaukee = data_Milwaukee$avg_temp_f,
  Chicago   = data_Chicago$avg_temperature_degrees_fahrenheit
) %>% unique()

ggplot(temperature_cities, aes(x = Milwaukee, y = Chicago)) +
  geom_point(alpha = 0.2) +
  geom_abline(intercept = 0, slope = 1, color = 'red') +
  labs(x = 'Milwaukee Average Daily Temperature (Degrees Fahrenheit)',
       y = 'Chicago Average Daily Temperature (Degrees Fahrenheit)',
       color = '') +
  facet_wrap(~year(date)) +
  theme_classic()
```



29. We can examine whether daily temperatures in Chicago and Milwaukee are drawn from a normal distribution using a Q-Q plot. Based on the Q-Q plot created from the code below, towhat extent are these temperature series normal? Try creating a similar Q-Q plot comparingAustin and Corpus Christi temperatures to the normal distribution. As an optional step, trycreating a Q-Q plot of Chicago average temperatures against Milwaukee temperatures. Arethese series drawn from the same distribution?

```
ggplot(temperature_cities %>% pivot_longer(!date, names_to = 'series', values_to ='temp'),aes(sample = temp, col
```

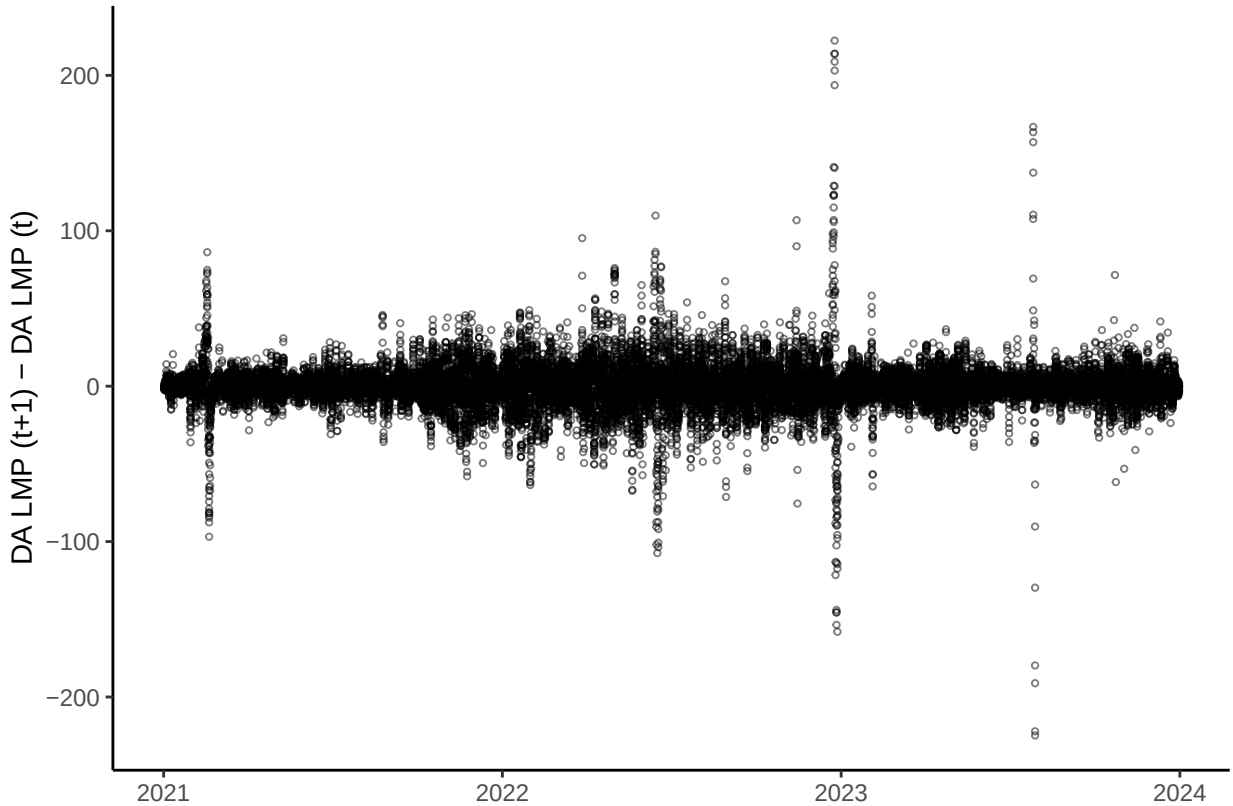


30. In February 2021, Texas suffered a devastating winter storm that exacerbated the vulnerabilities of its power grid. The code below displays average daily temperatures and real-time LMPs in Chicago, Milwaukee, and Corpus Christi during this month. Note the scalefor each plot. What do you observe from these plots? Conjecture as to the relationshipbetween prices and weather. Do you think the relationship is linear? Describe the role andnuances of anomalous weather.

VIII. Contracts for Differences

31. The code below plots day-over-day changes in the Chicago hourly DA LMP. What does eachpoint on the plot represent? Why are the observations on for most days approximatelysymmetric around zero? Which days exhibited the largest day-over-day changes in DA LMP?How do day-to-day changes in DA LMP relate to contracts for differences (CfDs)?

```
ggplot(data_Chicago[order(date)] %>%.[,prev_lmp_da := shift(total_lmp_da, type = 'lag'), by = hour],aes(x = date
```



32. Form your own hypotheses about how day-over-day changes in DA LMP may vary accordingto different dimension of the Chicago data (e.g., day-of-week, temperature changes, etc.).Test them. Discuss how patterns in day-over-day changes in DA LMP may relate to energystorage needs.

33. The code below creates a data set containing the ratio of day-over-day DA LMP spreads between Chicago and Milwaukee. That is, it computes the day-over-day change in hourly DA LMP for Chicago and Milwaukee and then computes their ratio in the spread_ratio field. Use this data set to plot spread_ratio and interpret the magnitude of spread_ratio. What do larger values mean and how do they relate to incentives for energy transmission and storage given your interpretation, what has changed in 2023 relative to 2021 and 2022?

```
internodal_spread <- data.table(date = data_Chicago$date,hour = data_Chicago$hour,day_of_week = data_Chicago$day
```

34. In the previous question, we explored the ratio of day-over-day DA LMP spreads between Milwaukee and Chicago. The code below creates a similar dataset for the ratio of day-overday DA LMP spreads between Chicago and Corpus Christi, TX; the spread_ratio_ercot variable holds this data (Corpus Christi belongs to the Electric Reliability Council of Texas,or "ERCOT"). What do larger values mean, and how do they relate to incentives for energy transmission and storage?

```
internodal_spread$spread_ratio_ercot <- internodal_spread$Chicago_da_spread /internodal_spread$CorpusChristi_da_
```

35. The code below tells you the number of days that spread_ratio and spread_ratio_ercot were above 1, respectively, between 2021 and 2023. Interpret these results.

```
count1 <- sum(internodal_spread$spread_ratio > 1, na.rm = TRUE)
count2 <- sum(internodal_spread$spread_ratio_ercot > 1, na.rm = TRUE)
print(count1)
```

```
## [1] 8860
```

```
print(count2)
```

```
## [1] 6082
```

```r
count3 <- sum(internodal_spread$spread_ratio > 0, na.rm = TRUE)
count4 <- sum(internodal_spread$spread_ratio < 0, na.rm = TRUE)
print(count3)
```

```
## [1] 16392
```

```r
print(count4)
```

```
## [1] 9792
```

36. Construct a spread metric that considers the contemporaneous spread between Chicago and Milwaukee to those from the preceding day. Interpret this ratio. How does it differ `fromspread_ratio`? Try comparing these two different series using a plot.

IX. Load and Price Duration Curves

37. Use the code below to load a dataset, created by Bolarinwa Ajanaku,23 containing daily observations of PJM data from January 1, 2016 to December 31, 2019.24 This dataset includes a number of series, including load, RT LMP, as well as supply and prices for natural gas and wind.

```r
daily_PJM <- fread(paste0(wd, "/", "PJM Daily.csv"))
```

38. The code below generates annual load duration curves for PJM, over the years 2016-2019.Based on the plot, which years featured higher demand for electricity? Try adapting the codeto annual price duration curves for PJM, using the rtm_price variable in place ofActual_load_GWh. Which year featured the highest LMP prices? Is this consistent with yourinference based on the load duration curve?

```r
# ldc <- daily_PJM[, .(Date, Actual_load_GWh)] %>%.[, Date := as.Date(Date, format = "%m/%d/%Y")] %>%.[, rank :=
#
# ggplot(ldc, aes(rank, Actual_load_GWh, color = factor(year(Date)))) +geom_line() +scale_x_continuous(expand =
```

39. Try plotting load duration curves for Austin, Milwaukee, and Corpus Christi, using the"load_mw" field. Which cities typically have higher loads?